

16	D	For zero resultant amplitude, two waves of the same type must meet in antiphase and have the same amplitude. When that happens, destructive interference occurs, and the resultant amplitude is the difference between the individual amplitude of the two waves. When two waves are emitted from their sources with the same intensity, they can travel different distances to meet at a point. Since intensity and hence amplitude decreases with distance from the source, they will meet with different amplitude. So Option A is incorrect. Two waves must be in antiphase with each other when they meet, so Option B is incorrect. Two waves do not need to travel in opposite directions for resultant amplitude to be zero. They can travel in any directions, as long as they fulfil the conditions stated in the first paragraph, to give zero resultant amplitude. So Option C is incorrect.	E
17	C		D